

CWT-based transfer learning model with optimal channel selection for the detection of MDD using EEG signals

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ABSTRACT

Major Depressive Disorder (MDD) is one of the most prevalent and severe psychiatric conditions that pose significant diagnostic challenges due to the complex and subtle nature of brain dynamics. Conventional EEG analysis methods often fall short in capturing these nuanced temporal-frequency dynamics. To address this limitation, we propose a deep learning framework that integrates Continuous Wavelet Transform (CWT), scalogram representations, and transfer learning with the VGG16 architecture. The rationale for employing CWT and scalograms lies in their ability to preserve both time and frequency information, enabling the extraction of clinically meaningful features of brain dynamics, while transfer learning with VGG16 leverages prior knowledge from large-scale image datasets, improving feature generalization and reducing the computational burden of training on limited EEG data. By combining these techniques with robust preprocessing and channel selection, the proposed model effectively identifies distinct spectral power alterations, particularly in theta and alpha bands, associated with MDD. The experimental results demonstrate that the proposed model outperformed traditional methods, achieving a more precise detection of MDD. These findings highlight the promise of combining advanced signal processing with deep learning for more reliable, non-invasive diagnosis of MDD through EEG signals, and support its clinical application.

1. Introduction

Major depressive disorder (MDD), also referred to as depression, is marked by persistent sadness, reduced interest in usual activities, cognitive impairments, and suicidal ideation. Depression is often accompanied by feelings of guilt, diminished self-esteem, poor concentration, and, in severe instances, thoughts of suicide [1]. The World Health Organization (WHO) reports that nearly 280 million individuals globally are affected by depression [2]. Although it can effectively be managed with psychotherapy and medication when properly diagnosed, depression remains a persistent public health concern as it frequently goes undetected. The diagnosis of MDD typically relies on an individual's self-reported experiences or behaviors observed by family or friends. This is usually supplemented by a mental status examination that follows standardized scale-based interview criteria, such as the Diagnostic and Statistical Manual for Depression (DSM-5) [3], Mini-Mental State Exam-

nation (MMSE) [4], Beck Depression Inventory (BDI) [5], and Hamilton Depression Rating Scale (HDRS) [6]. However, these approaches can be inconsistent as they exclude objective biomarkers in the diagnostic process. Furthermore, they heavily rely on patient cooperation, which can be compromised due to the societal stigma around sharing internal emotional distress.

Biomarkers like electroencephalogram (EEG) can be analyzed to develop a precise and consistent method for diagnosing MDD [7]. EEG recordings of brain activity reveal the relationship between mental states and neural activity, providing detailed insights into central nervous system processes [8]. EEG has been employed for mood recognition and can serve as a reliable biomarker for investigating mental and cognitive illnesses such as MDD [9].

Interpreting EEG signals manually is challenging due to their inherent heterogeneity, nonlinearity, and nonstationary nature. To address these issues, machine learning (ML) techniques have been

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integrated into computer-aided diagnosis (CAD) systems for diagnosing various complex neurological disorders, including dementia [10], sleep disorders [11–15], Parkinson's disease [16–19], schizophrenia [20], seizures [21–26], and epilepsy [27–30]. Several have been proposed for diagnosing MDD by analyzing EEG signals with the help of ML techniques [31–38].

Conventional ML methods for diagnosing MDD are often labor-intensive and ineffective due to the complexities and time-consuming nature of feature extraction from EEG signals. Additionally, these traditional ML techniques rely heavily on handcrafted features, which can result in decreased reliability. Adopting deep learning (DL) frameworks can mitigate these limitations by automating feature extraction and enhancing reliability.

In recent years, deep learning frameworks have emerged as powerful tools for diagnosing MDD using EEG signals. Several state-of-the-art architectures have been explored, including convolutional neural networks (CNN) [39], graph neural network (GNN) [40–42], deep hybrid models that combine CNN with Long-Short-Term Memory (LSTM) architectures [43]. Additionally, one-dimensional CNN (1D-CNN) and 1D-CNN-LSTM approaches have demonstrated notable performance in this domain [43]. The InceptionTime model and a strategic channel-selection approach have also been utilized to enhance classification accuracy [44]. As EEG signals contain much information on neural electrical activity, region-based analysis of EEG signals could play a significant role in identifying depression. Region-based analyses have shown added diagnostic value [45–48], yet are rarely integrated into advanced DL pipelines.

Due to the non-linear and dynamic nature of brain signals, many researchers have used non-linear techniques to analyze EEG signals of MDD patients [32,49,50]. Furthermore, non-linear techniques can effectively capture the chaotic patterns and abrupt shifts in EEG signals that arise from the inherent neurophysiological processes [51]. As a non-linear technique for extracting features from EEG signals, the continuous wavelet transform (CWT) based features have shown the potential to discriminate MDD patients from healthy controls [34,52–54]. The CWT enables better frequency resolution at low frequencies and improved temporal localization of transient phenomena in EEG signals within the time domain.

Recent developments in multimodal large language models (LLMs), which integrate interview audio, text, and physiological signals such as EEG, eye-tracking, or facial expressions, have introduced a novel transformer framework for the prediction and assessment of mental health conditions [55–58]. Though LLMs excel in recognizing linguistic patterns, they do not inherently understand the underlying psychological context or causality of mental disorders. Furthermore, the application of LLMs to depression detection poses several challenges, including data privacy and ethical concerns, limited model interpretability, domain bias, and cultural variability [59].

In this paper, we propose a framework that combines three innovations: (i) continuous wavelet transform (CWT) to generate time-frequency scalograms capturing the non-linear dynamics of EEG signals; (ii) transfer learning with VGG16, applied in both frozen and fine-tuned modes to exploit rich pretrained visual features without large training datasets; and (iii) region-specific analysis through multi-channel stacking and selective channel reduction to quantify the contribution of different brain regions. This integration of nonlinear representation, advanced feature extraction, and targeted brain-region analysis distinguishes our approach from prior learning-based solutions, enabling more precise and interpretable classification of MDD from EEG data.

2. Materials and methods

2.1. EEG dataset description

In this investigation, we employed a publicly available EEG dataset [34,60] to evaluate our proposed method of detecting MDD patients.

Table 1

Training and test segments for healthy controls and MDD patients.

Subjects	Training Sample		Test Sample		No. of Segments	
	EC	EO	EC	EO	Training	Test
Healthy	25	26	3	3	1017	120
MDD	28	28	2	3	1092	90

The EEG signals in the dataset were acquired by an EEG cap with 19 electro-gel sensors from both MDD and healthy control (HC) subjects in eyes-closed (EC) and eyes-open (EO) conditions, with each session lasting 5 min. The international 10–20 system [61] was followed for on-scalp placement of the sensors with linked-ear (LE) reference. The dataset comprises a total of 64 participants, 34 MDD patients (an average age of 40.3 ± 12.9), and 30 healthy control (HC) participants (an average age of 38.3 ± 15.6). The recorded EEG signals were sampled at 256 Hz, bandpass filtered between 0.1 and 70 Hz, along with a 50 Hz notch filter to suppress power line interference. Standard artifact removal procedures were applied to ensure signal quality.

2.2. Methodology

Our proposed methodology for detecting MDD patients employing multichannel EEG signals is illustrated in Fig. 1. The method encompasses several key steps: (i) preprocessing EEG signals, (ii) segmentation, (iii) channel selection, (iv) applying Continuous Wavelet Transform (CWT) to produce scalograms, (v) stacking multiple channel scalograms to form a single image, (vi) applying transfer learning with the VGG16 model (both frozen and unfrozen versions), and (vii) evaluating classification performance. Leveraging EEG data from 34 depressed and 30 non-depressed individuals, the model demonstrates its ability to discriminate EEG signals into depressive and healthy categories, showcasing significant improvements in diagnostic accuracy and reliability. All steps of our methodology are described in the following subsections.

2.2.1. Preprocessing EEG signals

EEG signals can often be contaminated with artifacts and noise during acquisition, originating from both biological and non-biological sources. These sources include blinking eyes, cardiac activity, movement of body parts, and muscle activity, as well as channel noise. As reported in [33], the EEG dataset employed in this study was preprocessed and verified to be free of significant artifacts. Consequently, no additional artifact removal procedures were applied. Instead, a bandpass filter ranging from 1 to 45 Hz was employed to retain the frequency components relevant to brainwave activities. This step ensures that our EEG data accurately represents brainwave activity, preventing erroneous analysis and preparing the data for subsequent steps such as segmentation, channel selection, and CWT analysis.

2.2.2. Epoching of brain signals

To facilitate comparative analysis and optimize precision, we divide the EEG signals into a fixed duration of 15 s, producing 20 segments for each subject. This segmentation approach ensures a consistent duration for each epoch, which is crucial for subsequent processing steps. We consider each segment as an individual subject for both EC and EO conditions. Consequently, this segmentation significantly increases the number of training and test instances (Table 1), which improves the robustness of the training process.

2.2.3. Channel selection

Fig. 2 illustrates the placement of 19 electrodes based on the well-recognized 10–20 system [62]. For this study, four distinct channel configurations were selected, each corresponding to a specific cortical region: right frontal (Fz, Fp2, F8, F4), left frontal (Fz, Fp1, F7, F3), central-parietal (C3, Cz, C4, P3, Pz, P4), and parietal (P3, Pz, P4). To identify the

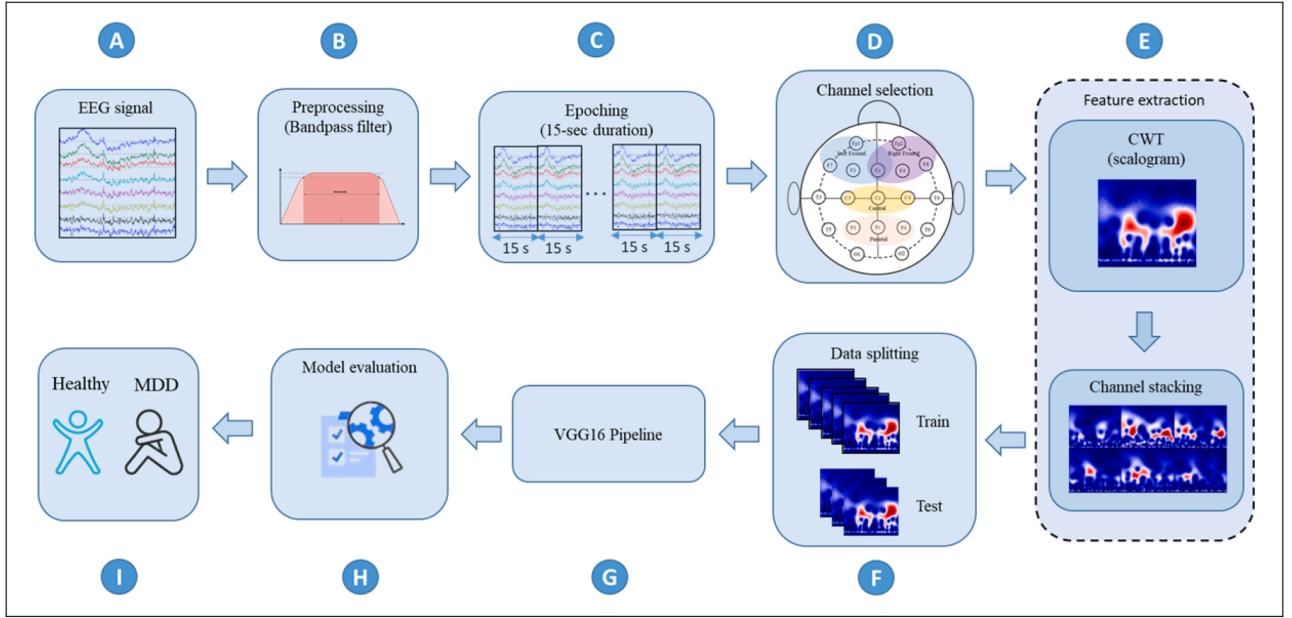


Fig. 1. Analytical pipeline of our proposed methodology for detecting MDD patients. A. EEG signals of MDD and healthy subjects obtained from 19 channels using the 10–20 system. B. Filtering the EEG signals with a bandpass filter within 1–45 Hz range. C. Segmentation of brain signals with 15-sec epoch duration. D. Channel selection based on four specific brain regions – right frontal, left frontal, central-parietal, and parietal nodes. E. CWT-based feature extraction and selected channel stacking. F. Splitting data into train and test sets. G. Transfer learning model using VGG16 pipeline. H. Model evaluation with test data. I. The test input was identified as either MDD or healthy.

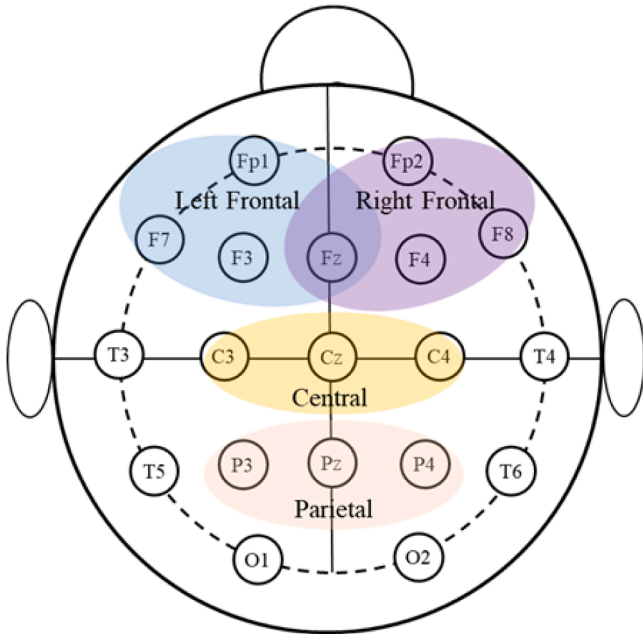


Fig. 2. Placement of 19 electrodes following the international 10–20 system. The shaded channels are selected for the proposed study.

optimal channel configuration, we applied the proposed method across the different sets of channels. The assumption behind this selection process is to capture relevant brainwave patterns specific to different brain regions, and that may provide a more accurate and reliable diagnostic result for MDD.

2.2.4. Feature extraction

Continuous Wavelet Transform (CWT) is an effective method for analyzing non-stationary signals, such as EEG. It decomposes the signal

into time-frequency space, capturing both the frequency content and its variation over time, which is crucial for understanding dynamic brain-wave activities. The CWT of a signal $x(t)$ is defined as

$$X_{\psi}(s, \tau) = \frac{1}{\sqrt{s}} \int_{-\infty}^{\infty} x(t) \psi\left(\frac{t-\tau}{s}\right) dt \quad (1)$$

where s represents the scaling parameter controlling the width of the wavelet, τ denotes the translation parameter controlling the shift of the wavelet, and ψ corresponds to the mother wavelet. In this study, the complex Morlet wavelet was employed as a mother wavelet.

In our methodology, we use the PyWavelets library to implement CWT. We apply CWT to the EEG signals of the selected channels to obtain wavelet coefficients. The absolute values of these coefficients represent the amplitudes of various frequency components at different times, which are then normalized to a suitable range for visualization. After normalization, we create a scalogram for each channel, which combines spectral information and the temporal structure of the EEG signals represented in the time-frequency plane as shown in Fig. 4. This scalogram provides a detailed insight into brainwave activities.

2.2.5. Channel stacking

Channel stacking is an essential step in our methodology, where we combine multiple channel scalograms to create a single image for each segment. We obtain scalograms for each selected channel after applying the CWT to the segmented EEG data. The scalograms of the selected channels are then stacked together to form a single composite image. For instance, if we select four channels from the left frontal region (Fz, Fp1, F7, F3), the scalograms of these four channels are combined into a single image. This combined image serves as a representative visualization of the segmented EEG data for that specific region. Each segment, now represented by this stacked channel image, is treated as a unique subject in the training process. The stacked images are resized to $224 \times 224 \times 3$ to be compatible with the VGG16 architecture.

Stacking multiple channel scalograms, we capture comprehensive information from different brain regions within a single image. This enhances the model's ability to learn and recognize patterns associated with MDD, leading to more accurate and reliable diagnostic outcomes.

2.2.6. Transfer learning with VGG16 model

In this step, we leverage the powerful capabilities of the VGG16 model through transfer learning to classify the EEG signals. Here is a breakdown of the process:

1. **Utilize the VGG16 Model:** We employ both frozen and unfrozen versions of the VGG16 model. Initially, the model is used with pre-trained weights.
2. **Freezing the Layers:** Up to the 10th layer, the VGG16 model's layers are frozen, i.e., the weights in these layers are not updated during the training phases, allowing us to retain the pre-trained feature extraction capabilities.
3. **Unfreezing the Layers:** Beyond the 10th layer, we unfreeze the VGG16 model. This enables the layers to be fine-tuned and updated during training, allowing them to adapt specifically to the characteristics of our EEG dataset.
4. **Data Augmentation:** To prevent overfitting, we apply data augmentation techniques such as random flips and rotations before feeding the images into the VGG16 pipeline. This enhances the diversity of the training dataset, helping the model generalize more effectively.
5. **Training with Stacked Channel Scalogram Images:** The stacked channel scalogram images, representing the EEG segments, are used to train the VGG16 model. This involves feeding the images into the model and optimizing the weights to minimize the classification error.

By freezing and then selectively unfreezing the layers, we strike a balance between leveraging the rich features learned by the VGG16 model on large-scale data and the specificity required for our MDD diagnosis task. Data augmentation further enhances the model's ability to generalize, leading to improved diagnostic precision and reliability.

3. Results

3.1. Brain topography analysis

Before performing the MDD patient detection process, we analyzed the CWT-based spectral power of EEG signals for HC subjects and MDD patients in the EC and EO states. As shown in Fig. 3, we found that the spectral power of EEG signals is higher in the EO state compared to the EC state for both HC and MDD patients. This corresponds to the understanding that brain activation is generally higher when the eyes are open. Furthermore, it is observed that the spectral power of the EEG signals is almost uniformly distributed over the different brain regions in both MDD and HC subjects in the resting state (EC), as shown in Fig. 3(A) and (C). In contrast, we observed that in the EO condition, power is predominantly concentrated in the brain's frontal region for MDD patients, as depicted in Fig. 3(D). Hence, we could interpret the fact that the frontal nodes might be crucial in diagnosing MDD.

3.2. CWT-based scalograms analysis

In this investigation, we utilize CWT to convert EEG signals into scalogram images, which illustrate the spectral power in the time-frequency domain, as depicted in Fig. 4. These scalograms of brain signals provide valuable insights into the various frequency bands that significantly contribute to the overall spectral power of the signals. As shown in Fig. 4, we observed that the power is mainly concentrated within the frequency range of approximately 3 to 14 Hz for the MDD patient in the EO state, irrespective of the four channels in the right frontal region, namely, Fz, Fp2, F8, and F4. This frequency range corresponds to the brainwaves delta (< 4 Hz), theta (≥ 4 Hz and < 8 Hz), and alpha (≥ 8 Hz and < 14 Hz). However, the spectral power for delta waves is found to be nominal compared to that of theta and alpha waves. In contrast, the most significant frequency band for healthy control subjects falls within the range of 4 to 25 Hz.

Table 2

The detection performances of the proposed method across different brain regions.

Brain Region	Precision	Recall	F1 Score	Test Acc.
Left Frontal	97.17	97.14	97.15	97.14
Right Frontal	99.53	99.52	99.52	99.52
Central-Parietal	93.28	92.38	92.42	92.38
Parietal	94.96	94.29	94.31	94.29

In the EC condition, no active frequency bands were detected in the subjects with MDD. In contrast, healthy control subjects exhibited a dominant frequency band similar to that observed in the eyes-open (EO) condition. Consequently, the CWT-based scalograms for the subjects with major depressive disorder differ significantly from those of the healthy control group, both for EO and EC conditions.

3.3. Brain region based classification performances

To prevent any potential data leakage, all experiments were conducted with strict subject-level separation. Initially, 10% of the total subjects (4 MDD and 3 HC) were set aside as an independent test cohort, ensuring that all EEG segments from these subjects were excluded entirely from the training and validation processes. This design guaranteed that no data from the same individual appeared across multiple subsets. The remaining 90% of the subjects (56) were employed in a 10-fold subject-wise cross-validation scheme, where each fold included mutually exclusive subjects for training and validation. Thus, model evaluation was consistently performed on unseen subjects, ensuring a fair and unbiased assessment of performance.

To improve reproducibility and robustness, the entire training-validation-testing pipeline was repeated using three different random seeds (7, 21, and 142). The final prediction results were obtained through soft voting across the 10 models generated for the seed value of 142.

The performance of the proposed model was assessed using five evaluation metrics: precision, recall, F1-score, test accuracy, and confusion matrix analysis across the four targeted brain regions. As presented in Table 2, the right frontal region (Fz, Fp2, F8, F4) yielded the highest performance across all evaluation metrics. The left-frontal region (Fz, Fp1, F7, F3) demonstrated comparable performance, indicating its strong discriminative potential as well.

Furthermore, the confusion matrices for the four selected brain regions, illustrated in Fig. 5, confirm the model's robustness. The results reveal that the proposed approach attains high sensitivity and specificity in differentiating MDD patients from HC subjects.

4. Discussion

In this work, we demonstrate a deep learning framework for MDD diagnosis using multichannel EEG signals, incorporating techniques such as CWT, scalogram generation, and transfer learning with the VGG16 model. We examined the effectiveness of various preprocessing steps, including filtering and segmentation, and employed data augmentation techniques to avoid overfitting. Our experimental evaluation revealed that segmentation is crucial in increasing the number of training and test instances, significantly enhancing the model's robustness and performance.

Using CWT scalograms extracted from EEG signals significantly improves the detection of MDD patients. Recently, an increasing number of researchers have begun to employ CWT for depression detection [34,52,53,63]. The CWT is a potent approach to quantifying brain dynamics by capturing both the frequency and temporal information of EEG signals. Additionally, the non-linear feature can capture the critical aspects related to brain dynamics [51]. From the scalograms of the EEG signals as shown in Fig. 4, it is evident that the CWT-based

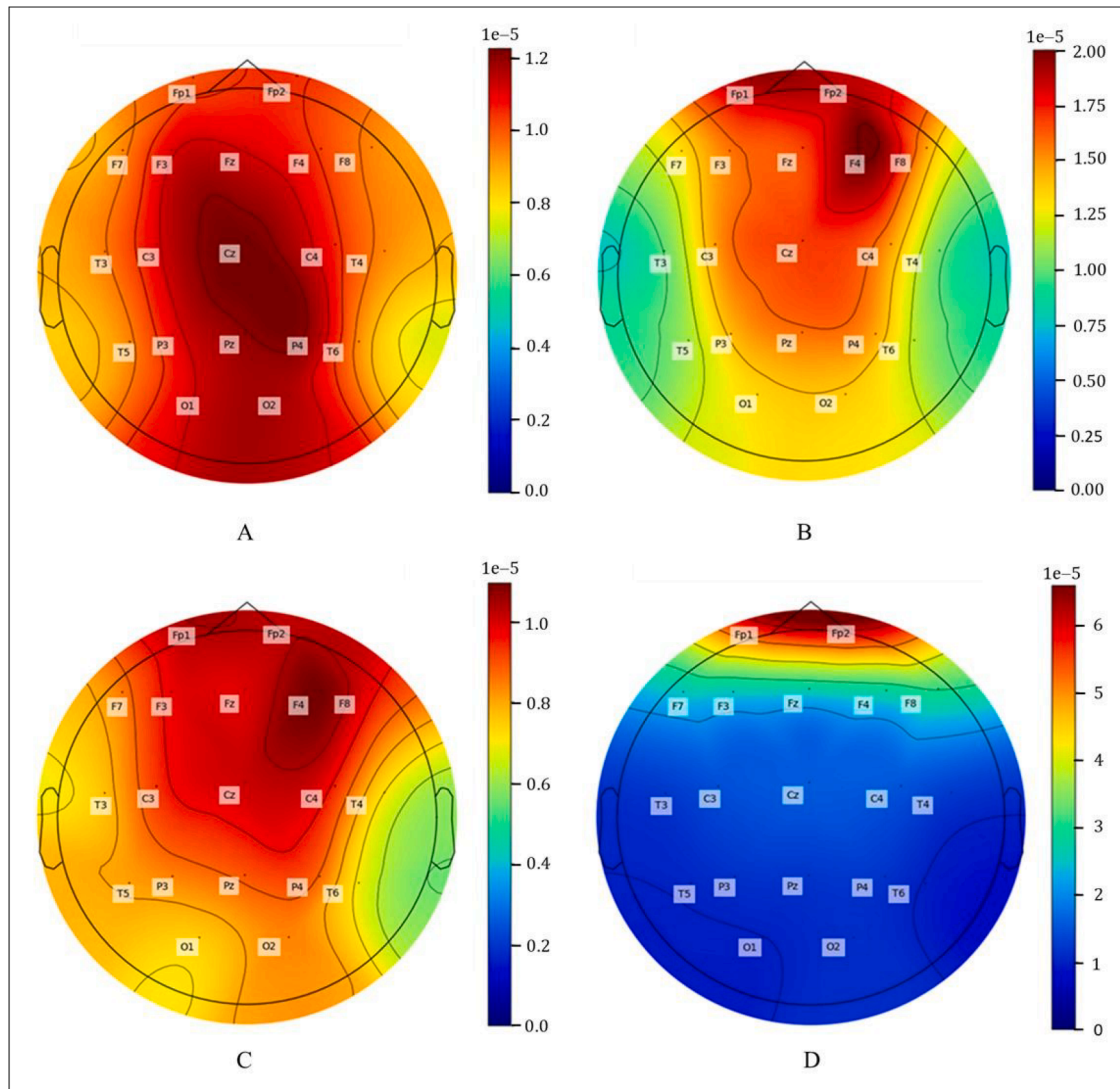


Fig. 3. Brain topography of the CWT-based spectral power. A. Healthy control (HC) in eyes-closed (EC) condition. B. Healthy control (HC) in eyes-open (EO) condition. C. MDD patient in EC state. D. MDD patient in EO state.

Table 3

The performance comparison of the proposed system with some recent studies.

Study	Classifier	Features	Data Size	Acc.
Proposed	VGG16	CWT Scalogram	34 MDD 30 HC	99.52
Mumtaz et al. (2017) [34]	LR	Wavelet Transform	34 MDD 30 HC	87.50
Mumtaz et al. (2017) [33]	SVM	Band power, α -asymmetry	33 MDD 30 HC	98.40
Mumtaz et al. (2018) [92]	SVM	Synchronization likelihood	34 MDD 30 HC	98.00
Movahed et al. (2021) [37]	RBFSVM	Statistical, spectral, wavelet, functional connectivity, nonlinear	34 MDD 30 HC	99.00
Saeedi et al. (2021) [93]	1DCNN-LSTM	Effective Connectivity (GPDC, dDTF)	33 MDD 30 HC	99.25
Yang et al. (2023) [46]	SVM	PSD, LZC	34 MDD 30 HC	92.40
Zhu et al. (2022) [42]	GICN	Functional connectivity	27 MDD 28 HC	96.50
Wang et al. (2024) [41]	GCTNet	Temporal and Frequency	34 MDD 30 HC	97.55
Wang et al. (2024) [94]	HGNN	Temporal	34 MDD 30 HC	93.50
Zhu et al. (2025) [95]	MTNet	EEG and Eye-tracking data	21 MD 28 HC	91.79
Wu et al. (2021) [96]	CK-SVM	Band power (BP), Coherence, HFD, KFD	200 MDD 200 HC	84.16

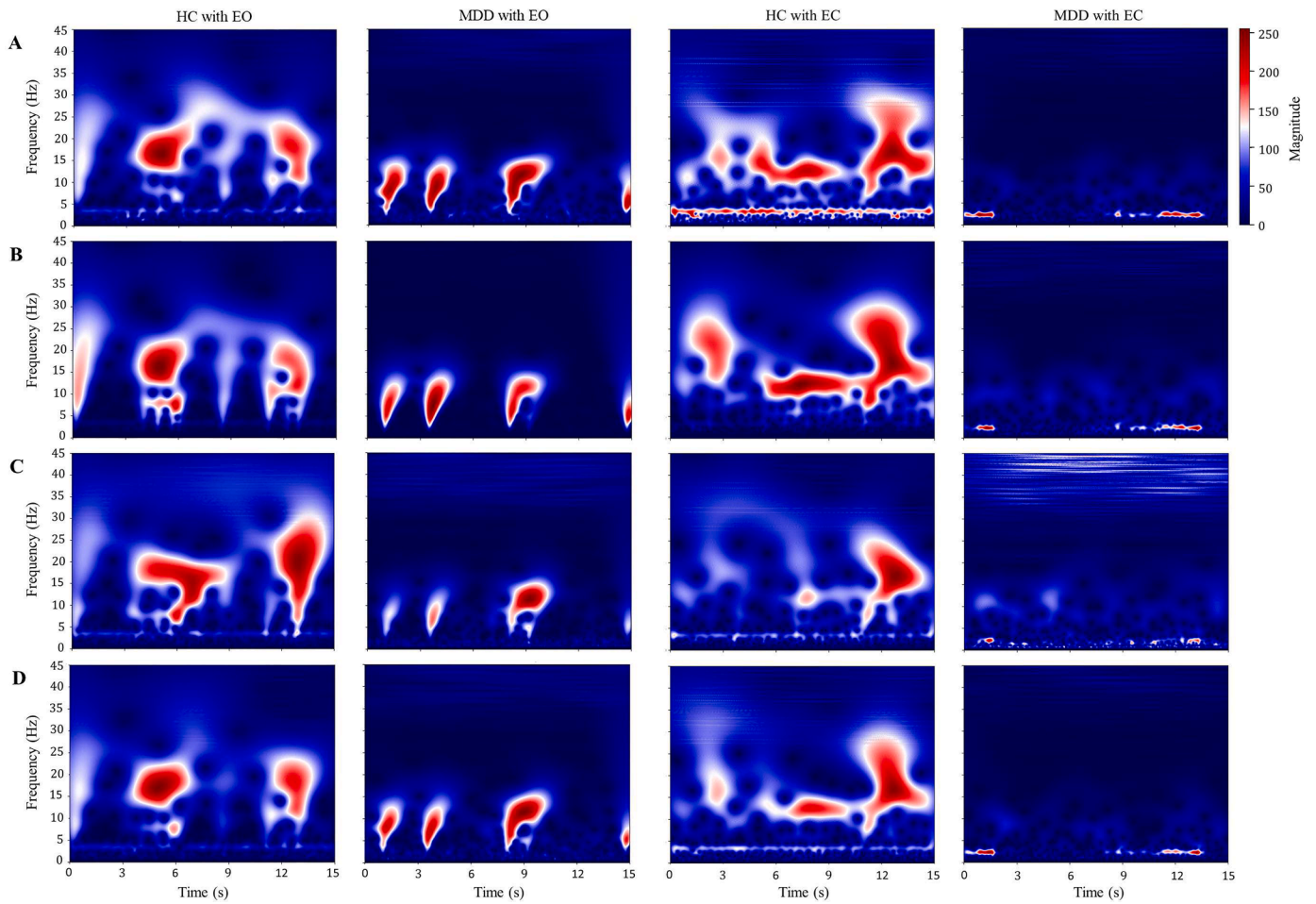


Fig. 4. Scalograms of the EEG signals for the right frontal region, both for HC and MDD subjects, in EO and EC states: A. Channel Fz, B. Channel Fp2, C. Channel F8, and D. Channel F4.

spectral power is predominantly concentrated at low frequencies within the range of 3 to 14 Hz for MDD patients in EO condition, while for healthy subjects, it is within the range of 4 to 25 Hz. More specifically, the MDD patients show higher activity in the theta-band (4 to 7 Hz) and alpha-band (7 to 13 Hz) under the EO condition compared to HC subjects.

Several studies have established a strong association between theta and alpha oscillations and memory-related processes. Elevated theta-band asymmetry in the frontal region may signify impairments in interhemispheric communication networks [64]. Alpha and theta activity are intricately linked to neural connectivity [65], memory processing [66,67], attentional control [68,69], and emotional regulation [70].

Recently, alpha-band activity has been reported to be the most prominent component in resting-state EEG, suggesting its potential as a biomarker [71]. Furthermore, several studies demonstrated the effectiveness of theta and alpha-band power in diagnosing MDD [37,72–74]. Based on the findings from the scalogram analysis, we suggest that theta and alpha oscillations may play a significant role in the pathogenesis of MDD. Consequently, the scalogram-based detection process can effectively discriminate MDD patients from healthy controls.

We applied the proposed method across different groups of channels to identify the optimal configuration for diagnosing MDD. Then, we compared the classification performance of the models trained on four specific brain regions: the right frontal region, the left frontal region, the central-parietal regions, and the parietal region.

The selection of frontal (right/left), parietal, and central regions was guided by previous evidence highlighting their relevance to MDD. The

frontal alpha asymmetry has long been considered a potential biomarker for depression [75–79]. Contributions of parietal cortical activities to the attentional networks and the default-mode network are frequently altered in depressive individuals, reflecting deficits in attention and self-referential processing [80,81]. Moreover, altered central (sensorimotor) activity and connectivity patterns have been associated with psychomotor retardation and disrupted thalamo-cortical communication in MDD [82,83]. Therefore, these regions were selected to capture affective, cognitive, and sensorimotor signatures that collectively characterize MDD.

Our findings revealed that the best performance was achieved for the right frontal region comprising channels Fz, Fp2, F8, and F4, resulting in a test accuracy of 99.52% (Table 2). In support of this result, we found that the CWT-based spectral power is mainly concentrated in the brain's frontal region for MDD patients in both EC and EO conditions (Fig. 3). It has been reported that the frontal region is the most common region to manifest anatomic abnormalities in MDD [84]. In the case of the left frontal region, we have a comparable accuracy (97.14%) with the right frontal region, indicating its potential in diagnosing MDD.

According to Li et al. [85], the strength of the functional connectivity of the frontal regions of MDD patients is higher than that of other brain regions. Impaired cognitive function in MDD has also been associated with frontal lobe dysfunction [86]. In addition, several studies have proposed that frontal alpha and theta asymmetry may serve as potential biomarkers for MDD [87,88]. Flor-Henry et al. [89] reported that depression is linked with relatively increased activity in the right frontal cortex compared to the homotopic region in the left hemisphere. Consistent with this, Hecht [90] suggested that depression involves

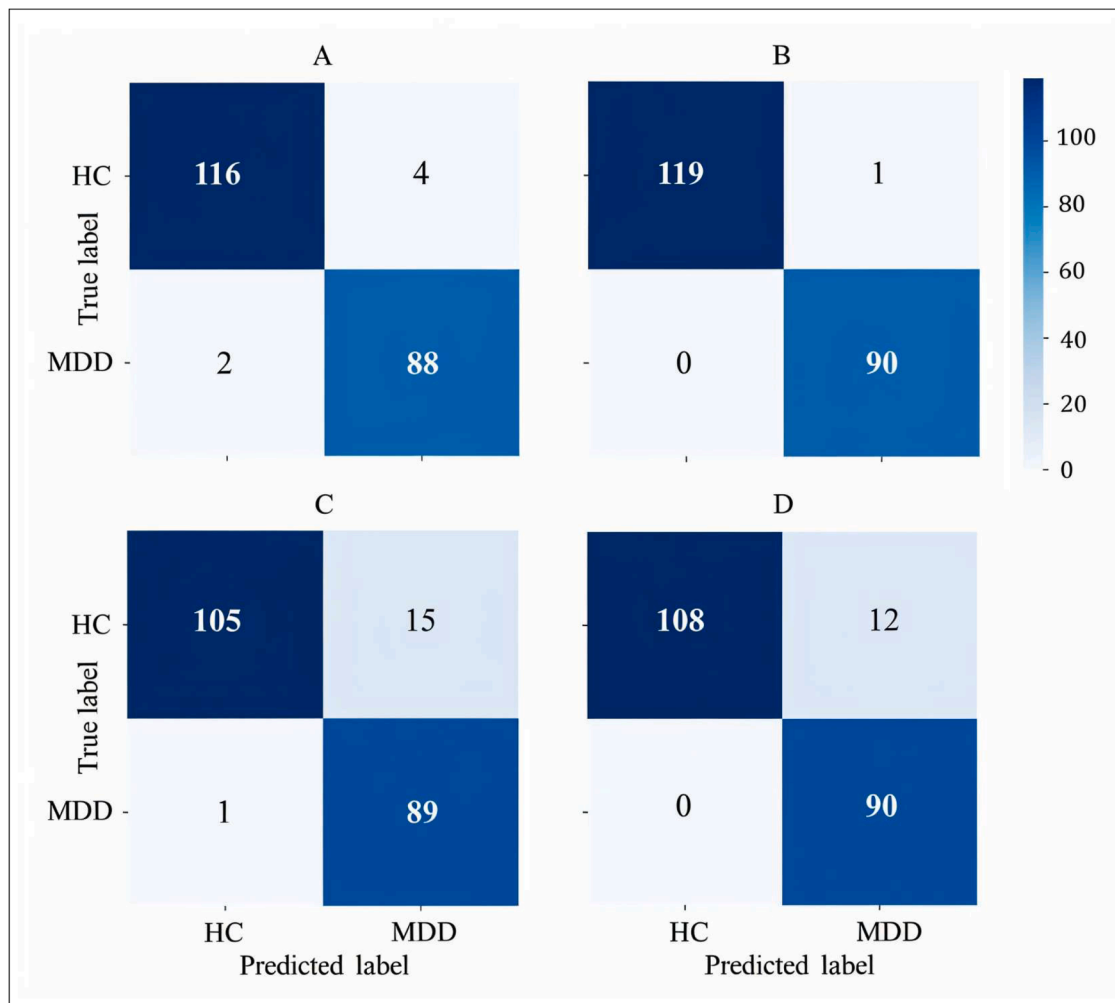


Fig. 5. Confusion matrix for four selected brain regions. A. Left frontal region, B. Right frontal region, C. -Central-Parietal regions, and D. Parietal region.

interhemispheric imbalance, reflected by hyperactivity in the right hemisphere and hypoactivity in the left.

Therefore, by focusing on robust preprocessing, essential channel selection, and advanced classification techniques, the proposed approach demonstrates superior performance in accurately identifying MDD patients using EEG signals, thus laying the foundation for more reliable and effective diagnostic tools. Although the frontal nodes showed strong discriminative potential for MDD classification in this study, it is important to consider that MDD reflects dysfunctions across multiple interconnected brain networks, including the default mode network, salience network, and central execution network, which together contribute to the disorder's clinical manifestations [91].

Several studies have employed machine learning techniques that leverage various features based on EEG signals to differentiate patients with MDD from healthy controls, as outlined in Table 3. Mumtaz et al. [34] proposed a machine learning model using logistic regression (LR) with wavelet decomposition of EEG signals as features, obtaining an accuracy of 87.5%. This study included 34 MDD patients and 30 healthy individuals. In a subsequent study, Mumtaz et al. [33] developed a machine learning scheme that employed several classifiers (LR, NB, SVM). This approach utilized EEG band power and alpha asymmetry, resulting in an improved accuracy of 98.4% for the SVM classifier. Later, Mumtaz et al. [92] introduced another machine learning framework that used synchronization likelihood features to detect MDD patients, achieving an accuracy of 98% for the SVM classifier. In a related study [37], researchers reported a remarkable accuracy of 99% by implementing a

machine learning framework that extracted features through various methods, including statistical, spectral, wavelet, functional connectivity, and nonlinear analysis techniques, where the sequential backward feature selection (SBFS) algorithm was employed to reduce the dimensionality of the feature space. In [93], a combination of CNN (1D and 2D) with an LSTM model was proposed in which effective brain connectivity was extracted by Generalized Partial Directed Coherence (GPDC) and direct Directed Transfer Function (dDTF) methods, achieving an outstanding accuracy of 99.25% with a very high computational cost. Yang et al. [46] introduced a nonlinear feature-based SVM classifier that utilized EEG signals from multiple brain regions in 34 individuals with MDD and 30 healthy controls, achieving an accuracy of 92.4%.

Graph neural networks have emerged as the leading architecture for representing brain functional networks in recent years. In a study by Zhu et al. [42], the researchers introduced the Graph Input Layer Attention Convolutional Network (GICN). This model incorporates a trainable weight matrix in the input layer and utilizes the brain functional network as its adjacency matrix. Through a 10-fold cross-validation, the GICN model achieved an impressive classification accuracy of 96.50%, surpassing many existing approaches. More recently, Wang et al. [41] presented GCTNet, which combines graph convolutional networks and transformer networks. This model integrates residual graph convolutional networks (GCN) with transformer blocks to effectively capture spatial-temporal EEG features, achieving a test accuracy of 97.55%. Additionally, in another study by Wang et al. [94], a hybrid graph neural network (HGNN) was proposed for classifying major depressive

disorder (MDD), resulting in an accuracy of 93.50%. Zhu et al. [95] designed a Multimodal Transformer Network (MTNet) for the detection of mild depression (MD) by integrating EEG and eye-tracking data from 21 cases and 28 healthy controls, achieving a recognition rate of 91.79%.

Wu et al. [96] presented a large and diverse dataset of resting-state EEG signals of 400 participants (200 MDD patients and 200 healthy controls) for MDD detection. This study utilized four EEG features, namely band power (BP), coherence (COH), Higuchi's fractal dimension (HFD), and Katz's fractal dimension (KFD), to train the SVM with a conformal kernel (CK-SVM), achieving an accuracy of 84.16%.

The proposed model shows promising performance in identifying MDD; however, its generalizability is limited due to the relatively small size of the dataset. To address this issue, future research should concentrate on strategies to expand both the size and diversity of the dataset, as well as improve the model's robustness and generalizability.

5. Conclusion

Our proposed model has shown superior performance compared to state-of-the-art models in the diagnosis of MDD. By leveraging techniques like CWT, scalograms, and transfer learning with the VGG16 model, we achieved high accuracy and reliability in classifying MDD patients from EEG signals. What sets our model apart is its unique capability to utilize multiple channel scalogram images and strategically select channels, such as the right frontal nodes (Fz, Fp2, F8, F4), which have shown exceptional performance in terms of both accuracy and F1 score. This feature could be particularly beneficial for clinicians, providing a robust and efficient tool for diagnosing MDD.

By incorporating data augmentation techniques such as random flipping and random rotation, we ensured that our model generalizes well, reducing the risk of overfitting. Our comprehensive evaluation confirmed the model's robustness and effectiveness, demonstrating significant improvements in diagnostic accuracy and reliability. Although we conducted our tests using a limited publicly available dataset, we remain optimistic that additional data collection can further enhance our model's performance. This study highlights the potential of advanced deep learning approaches to improve the precision and accessibility of MDD diagnosis, laying a promising foundation for more reliable and effective diagnostic tools in clinical practice.

6. Contributions

All authors contributed to the manuscript. R.M., M.B.I., and M.A.M. conceived and designed the study. R.M. analyzed the data and wrote the Python code to develop the pipeline. R.M., S.A.M., and M.B.I. conducted all other analyses and wrote the manuscript. M.A.M., F.B.S., S.A., and T.J.A. revised the manuscript. All authors read and approved the final version of the manuscript.

7. Competing interests

The authors declare that they have no conflict of interest.

CRediT authorship contribution statement

Rudro Mohanto: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization; **Md. Shamim-Al-Mamun:** Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization; **Farhana Binte Sufi:** Writing – original draft, Supervision, Project administration, Investigation, Data curation, Conceptualization; **Sarwar Ali:** Writing – original draft, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization; **Tahani Jaser Alahmadi:** Writing – review & editing, Resources, Project administration, Investigation; **Mohammad Ali Moni:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding

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Data availability

Data will be made available on request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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